

AeroRescue Nexus

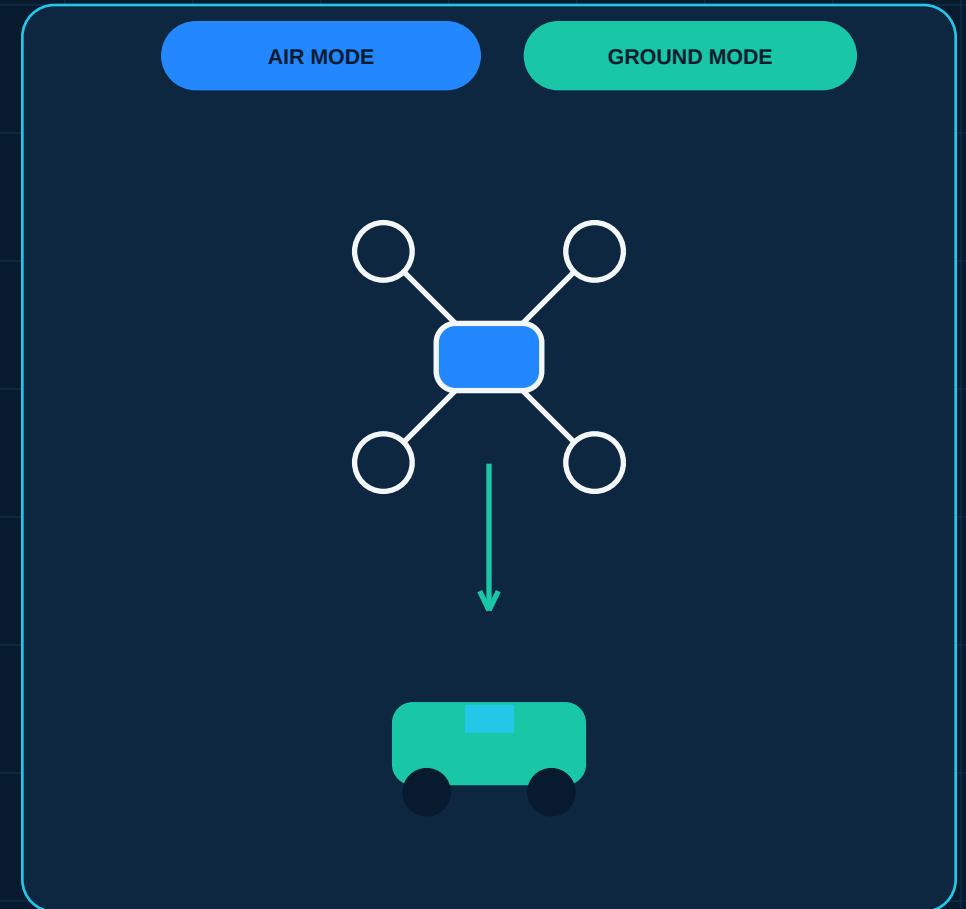
Autonomous Disaster Response UAV + Ground Rover System

A multi-mode rescue platform for survivor detection, disaster mapping, resilient communication, risk assessment and emergency assistance.

DESIGN TARGET

AI-first • Human-supervised • Resilient

Concept target — not a guaranteed field-performance claim.



01 | Problem Alignment

College source + proposed extension

COLLEGE PROBLEM STATEMENT

#47 — Disaster Response Drone for Remote Areas

Disaster response in floods, cyclones, earthquakes and landslides. Remote or blocked locations make rescue access difficult. Core requirement: aerial monitoring, victim location, GPS and emergency support.

AERORESCUE NEXUS EXTENSION

Adds an air + ground dual-mode rescue architecture.

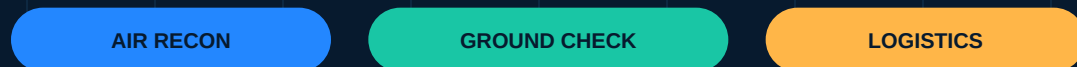
Adds resilient multi-UAV communication and task allocation.

Adds 3D situational mapping, risk scoring and autonomous logistics.

SYSTEM STORY



The project is not just a drone: it turns aerial observations into prioritized, actionable rescue missions.



Thermal/radar improve evidence; neither is guaranteed through-rubble vision.

02 | System Architecture

Five cooperating layers

1 SENSING

RGB + thermal + depth/range + IMU + GPS + environmental sensors

2 EDGE AI

Detection, tracking, mapping, risk scoring, mission prioritization

3 RESILIENT COMMS

LoRa relay / mesh + local telemetry + store-and-forward

4 AUTONOMY

Search planning, task allocation, energy management, failover

5 COMMAND

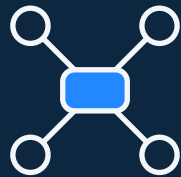
Map, alerts, evidence timeline, human approvals and overrides

Design principle: autonomy performs routine work; humans approve high-impact actions and can take control when confidence or system health drops.

03 | Dual-Mode Rescue Platform

Air reconnaissance + ground reconnaissance

AIR MODE • UAV



- Rapid wide-area search and live aerial assessment.
- AI vision + thermal anomaly cues + GPS geotagging.
- 3D mapping and blocked-route assessment.
- Acts as a communication relay when needed.

GROUND MODE • ROVER



- Deploy when aerial view is obstructed or ambiguous.
- Low-speed reconnaissance around debris and structures.
- Close-range camera / range sensing for verification.
- Return to a safe point when autonomy or battery is low.

AI RECOMMENDS → SAFETY CHECKS → SWITCH

04 | 30-Function Capability Set • 1/2

Professionally built for you

ID	Capability	Purpose
1	AI visual survivor detection	Find visible people in live/recorded imagery.
2	Thermal anomaly detection	Flag surface heat anomalies as supporting evidence.
3	Motion / micro-motion analysis	Analyze movement / micro-motion cues.
4	Radar vital-sign proxy (where feasible)	Use radar-derived vital-sign proxy where feasible.
5	GPS survivor geotagging	Attach coordinates to candidate detections.
6	Live video & telemetry	Stream video, position and telemetry.
7	Rapid 3D disaster mapping	Build rapid 3D scene / orthomosaic.
8	Debris / obstacle classification	Classify debris and blocked passages.
9	Structural instability indicators	Estimate structural-risk indicators from observed cues.
10	Safe-route scoring	Score safer movement corridors.
11	Air-to-air LoRa relay	Extend telemetry through UAV-to-UAV relay.
12	Adaptive mesh / link-quality routing	Route around weak links.
13	Network-loss fallback	Store data during network loss and forward later.
14	Autonomous search-grid planning	Generate and update search grids automatically.
15	Survivor priority scoring	Rank candidate locations by urgency and evidence.

DESIGN LOGIC Perception → evidence fusion → confidence → location → recommended action

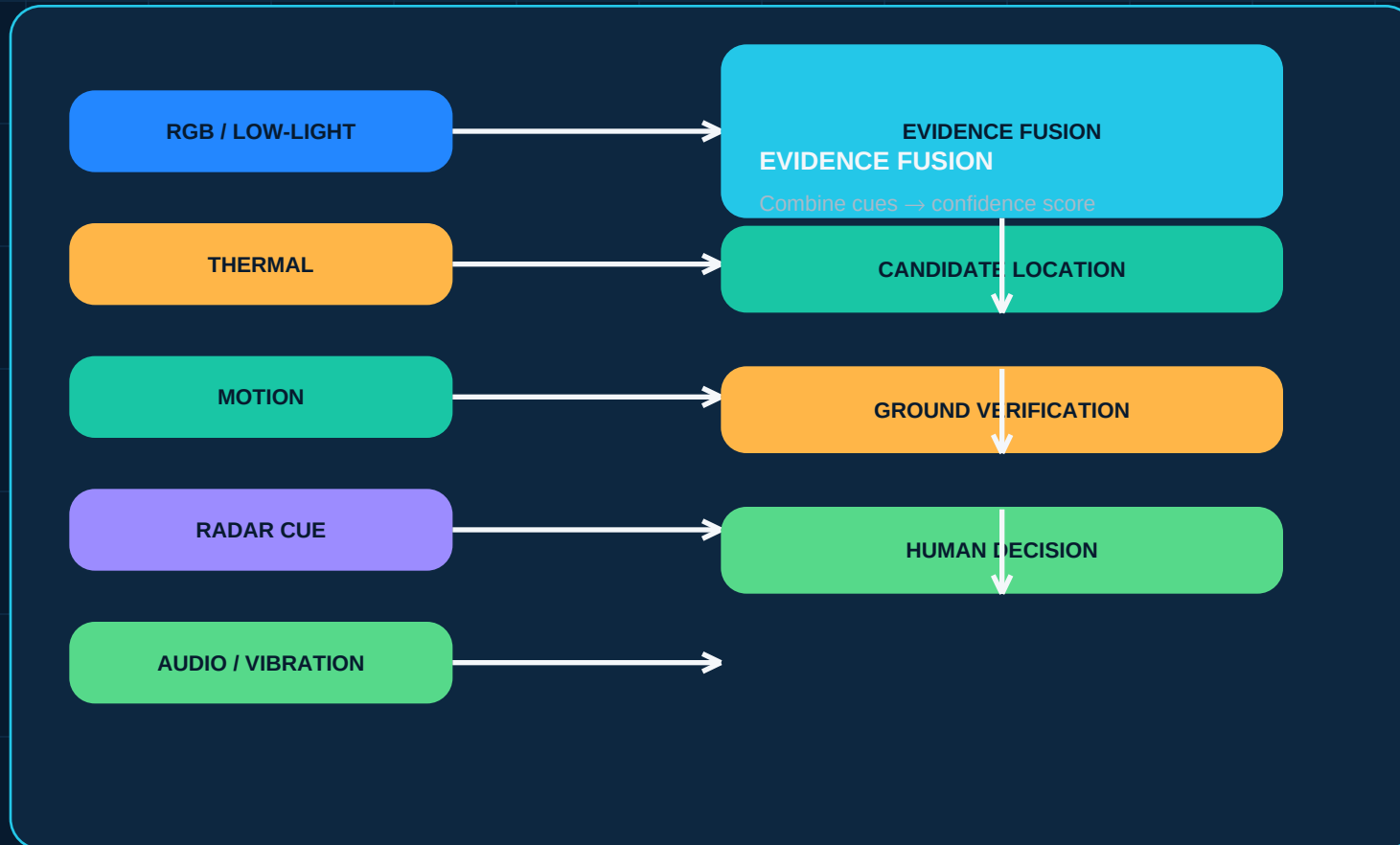
05 | 30-Function Capability Set • 2/2

ID	Capability	Purpose
16	Multi-UAV task allocation	Choose a unit by availability, energy, distance, link and mission.
17	Drone availability/state management	Track AVAILABLE / ASSIGNED / DELIVERING / CHARGING / FAULT.
18	Autonomous landing / takeoff assistance	Support safe transition with checks; human approval for high-risk actions.
19	Solar-assisted charging	Use solar as charging assistance, not as guaranteed in-flight power.
20	Battery health & energy prediction	Estimate remaining mission time and preserve reserve energy.
21	Auto return on inactivity / low battery	Return when inactivity, low battery or link loss crosses a policy threshold.
22	Human override & permission gates	Human approval for mission start, payload release and unsafe situations.
23	AI-to-human failover	Escalate when AI confidence or health falls below threshold.
24	Medical / food payload delivery	Carry compact first-aid, water, food or emergency beacon payloads.
25	Dynamic delivery-drone assignment	Choose the best delivery unit dynamically; no permanent pairing.
26	Delivery confirmation / payload status	Confirm delivery and payload state before return.
27	Air + ground rover dual mode	Switch from aerial assessment to ground verification.
28	Ground-rover indoor / rubble reconnaissance	Inspect obstructed spaces at low speed with close-range sensing.
29	Mission evidence log & event timeline	Keep timestamp, sensor, confidence, GPS, unit and operator actions.
30	Command dashboard & alerts	Present map, fleet state, alerts, evidence and approvals in one interface.

MISSION LOOP Detect → prioritize → assign → deliver → confirm → return → recharge → available

06 | Survivor Detection Stack

Sensor fusion instead of one-sensor certainty



ENGINEERING LIMIT

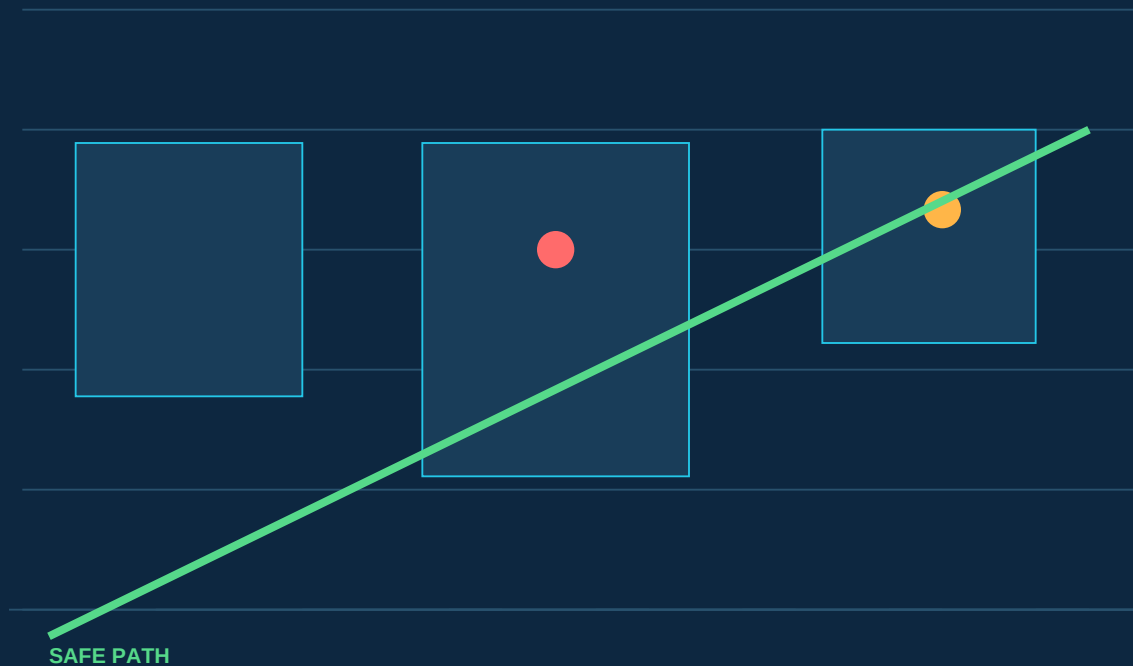
Thermal cameras generally detect surface heat signatures; solid concrete or rubble can block or distort them. A heartbeat through rubble is not guaranteed.

SAFE SIH CLAIM

- Detect visible people and thermal anomalies.
- Use motion/radar cues where supported.
- Flag ambiguous locations for rover or human verification.
- Report confidence and evidence — not an alive/dead binary claim.

07 | 3D Disaster Mapping

From camera frames to a navigable scene



PIPELINE

- Capture overlapping RGB frames during controlled flight.
- Estimate camera pose and reconstruct a 3D scene / orthomosaic.
- Overlay survivor markers, blocked roads, safe corridors and hazard zones.
- Let the operator rotate and inspect from multiple viewpoints.

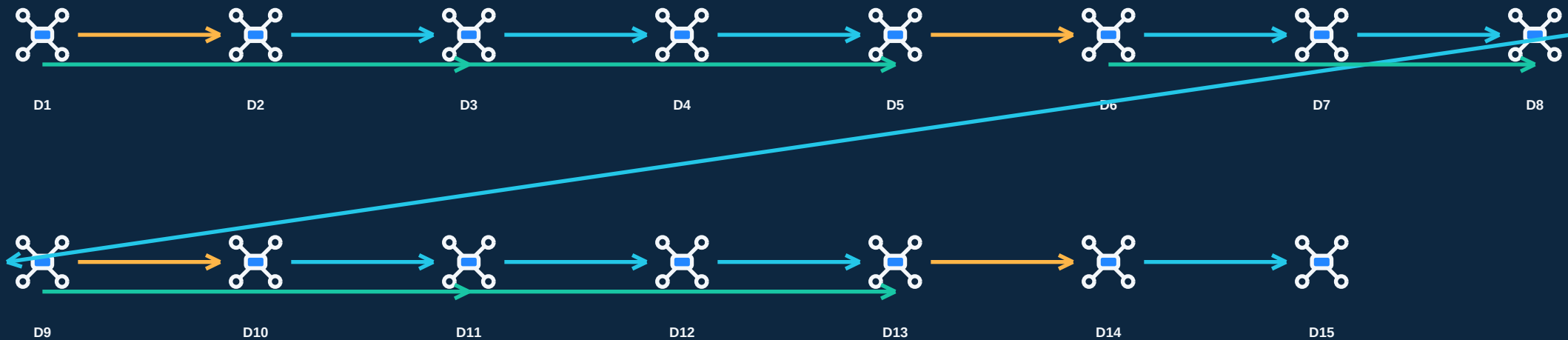
SURVIVOR PIN

HAZARD

SAFE PATH

08 | Resilient Multi-UAV Communication

D1 → D2 → ... → D15 relay concept with adaptive routing



LINK LOGIC

Each UAV measures link quality. If a hop is weak/unavailable, choose another reachable relay or store data until the link returns.

RANGE NOTE

Do not promise fixed 50 m / 10 km LoRa range; validate radio, antenna, terrain, frequency rules and obstacles experimentally.

09 | 75% AI / 25% Human — Supervised Autonomy

Automation with explicit safety gates

AI • ≈75% ROUTINE WORK

- Search planning
- Detection + tracking
- Task allocation
- Energy / availability
- Routine return

HUMAN CONTROL / APPROVAL

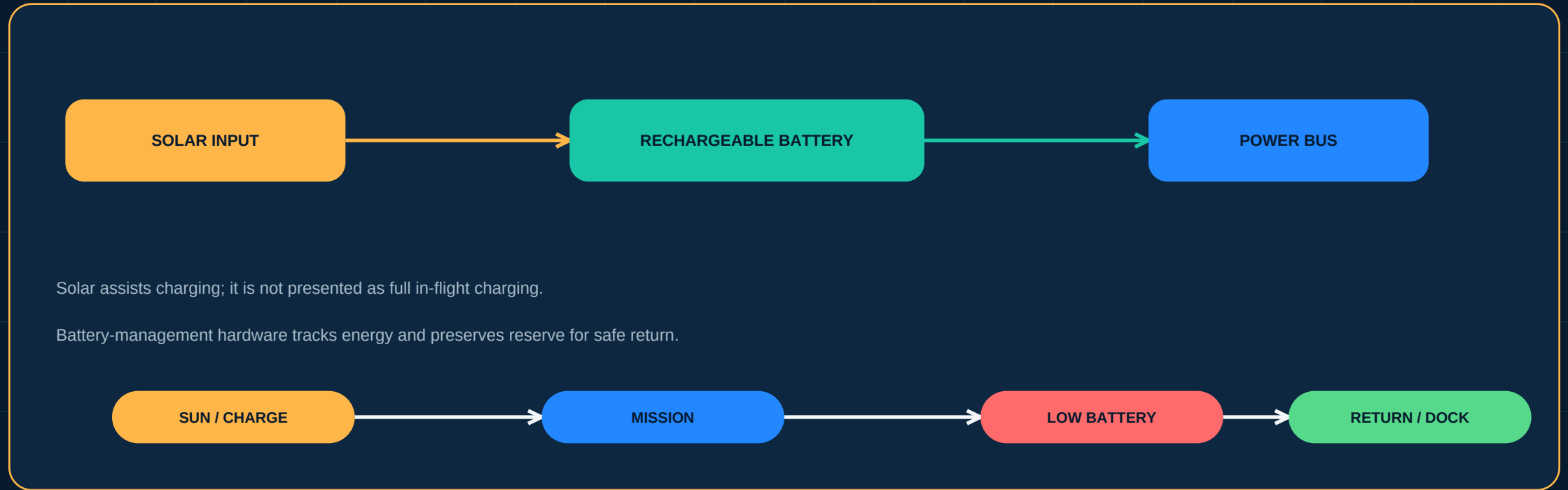
- Mission start / high-risk permissions
- Manual takeover
- Payload release authorization
- Unsafe landing override
- Mission termination
- AI-failure response

AI HEALTH → CONFIDENCE → ESCALATE

Escalate when AI confidence, navigation health, communication health or battery safety falls below threshold.

10 | Energy System

Solar-assisted charging + rechargeable battery



ENERGY POLICY Mission time prediction + reserve energy + auto return + dock/recharge + return to AVAILABLE state.

11 | Fleet Availability + Failover

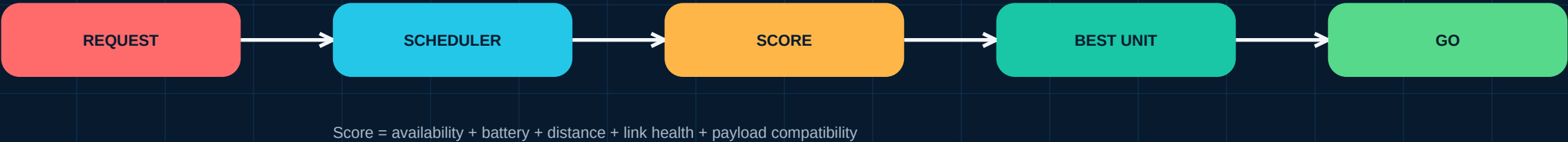
No single delivery drone is permanently assigned

STATE	MEANING	SCHEDULER ACTION
AVAILABLE	Ready for a new mission	Eligible
ASSIGNED	Reserved for a mission	Monitor
DELIVERING	Payload mission in progress	Unavailable
CHARGING	Energy recovery	Wait / re-score
FAULT	Health or link issue	Exclude + alert



EXAMPLE

D16 is delivering and becomes unavailable. The scheduler checks D17–D30 and chooses the best available unit.



12 | Emergency Assistance Loop

Food / water / medical-kit delivery as a controlled function

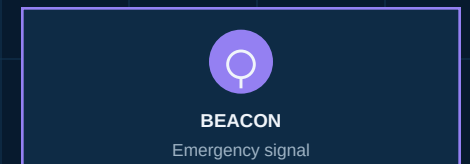
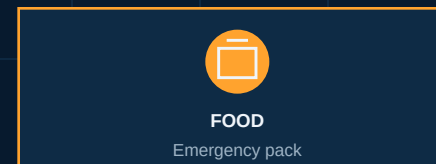
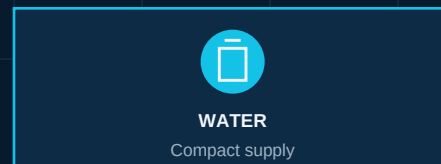


PAYLOAD EXAMPLES

First-aid kit • water • compact food pack • emergency beacon

Prototype rules should enforce payload weight, safe drop/landing rules and human authorization for release. Delivery is assistance, not medical treatment; do not claim AI alone can determine whether a detected person is alive or dead.

CONTROLLED PAYLOAD OPTIONS



13 | Rescue Command Dashboard

One screen for evidence, maps, fleet and alerts

SURVIVORS

03

UAVs ACTIVE

12

DELIVERY READY

07

LINK HEALTH

92%

BATTERY RESERVE

31%

EVIDENCE RECORD

- Timestamp
- Sensor source
- Confidence
- Coordinates
- Assigned unit
- Operator actions

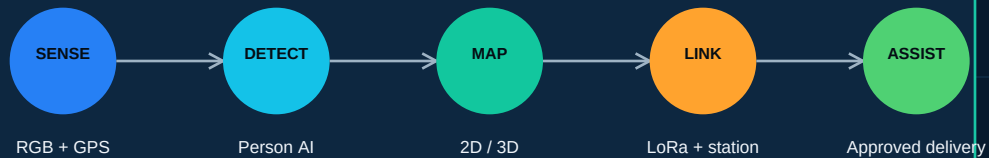
Dashboard connects live 2D/3D maps, fleet state, alerts, evidence timeline and human approvals

14 | Feasibility, Risks & MVP

Build the high-value core first; keep advanced functions modular

MVP — REALISTIC DEMO

MVP BUILD PATH



Each stage remains human-supervised before high-impact actions.

- One UAV + one ground rover
- RGB + GPS + onboard compute
- AI person detection on recorded/live feed
- Basic 3D/2D mapping
- LoRa telemetry with a ground station
- Human-approved delivery simulation
- Automatic return on low battery / lost link

MAIN DIFFICULTIES + MITIGATION

RISK ? MITIGATION MAP



LoRa range varies → validate radio, antenna, terrain and obstacles experimentally.

3D mapping needs overlap, compute and positioning quality → constrain the demo area.

Autonomous landing near rubble is difficult → use safe marked landing zones in MVP.

Solar adds mass and limited energy → treat it as charging assistance.

Multi-UAV autonomy is much harder → simulate fleet orchestration before scaling hardware.

RECOMMENDED CLAIM Target $\geq 95\%$ on a defined test dataset / scenario — not 95% guaranteed in all disasters.

15 | Differentiation + References

Strong architecture is more defensible than one flashy sensor

DIFFERENTIATION STACK

01 Air + ground mode

02 Multi-sensor evidence fusion

03 Adaptive UAV relay network

04 Dynamic search → delivery fleet

05 AI-first + human safety gates

06 Energy + availability orchestration

ORIGINALITY POSITIONING

Thermal rescue drones, UAV mapping and UAV relay networking already exist. Do not claim those individual functions are “never done before.” The defensible novelty is the integrated architecture and your validated implementation.

REFERENCE CONTEXT IN THE BLUEPRINT

- DJI Enterprise — aerial thermal search & rescue
- Pix4D — rapid UAV disaster mapping
- MDPI Electronics — UAV gateway / LoRaWAN networking
- U.S. DOT / NTL — UAS disaster-response applications
- ICAS 2024 — thermal-sensor UAVs for rubble search

FINAL POSITION AeroRescue Nexus = #47 base problem + integrated multi-mode rescue orchestration.